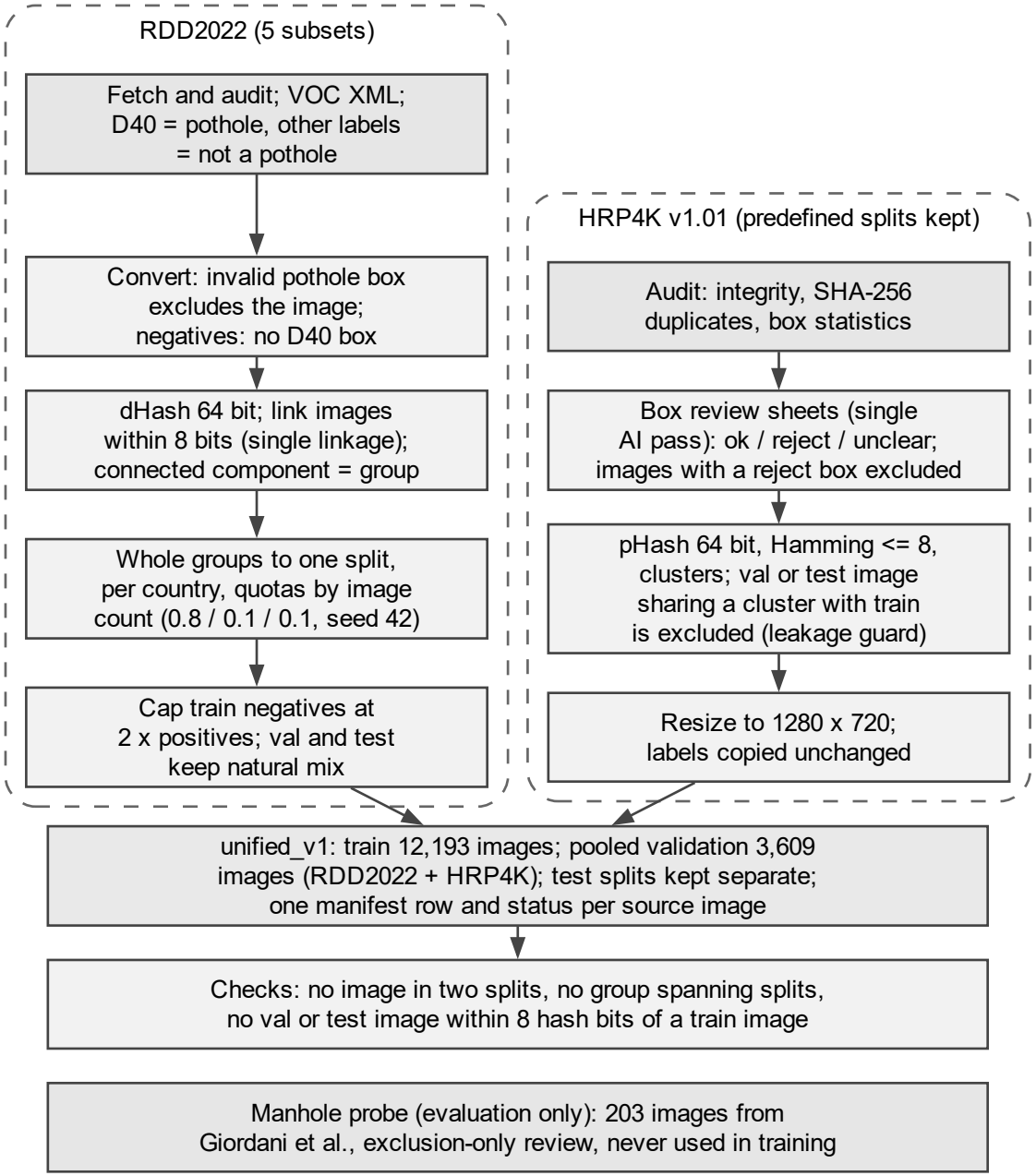
RDD2022 (5 subsets)
Fetch and audit; VOC XML; D40 = pothole, other labels = not a pothole
Convert: invalid pothole box excludes the image; negatives: no D40 box
dHash 64 bit; link images within 8 bits (single linkage); connected component = group
Whole groups to one split, per country, quotas by image count (0.8 / 0.1 / 0.1, seed 42)
Cap train negatives at 2 x positives; val and test keep natural mix
HRP4K v1.01 (predefined splits kept)
Audit: integrity, SHA-256 duplicates, box statistics
Box review sheets (single AI pass): ok / reject / unclear; images with a reject box excluded
pHash 64 bit, Hamming <= 8, clusters; val or test image sharing a cluster with train is excluded (leakage guard)
Resize to 1280 x 720; labels copied unchanged
unified_v1: train 12,193 images; pooled validation 3,609 images (RDD2022 + HRP4K); test splits kept separate; one manifest row and status per source image
Checks: no image in two splits, no group spanning splits, no val or test image within 8 hash bits of a train image
Manhole probe (evaluation only): 203 images from Giordani et al., exclusion-only review, never used in training